Odd kernels, cyclotomic modules and cyclic carries

The human_compression project

Topic F: expanded common theory, private draft

Abstract

We combine the complete existing odd-kernel and cyclotomic lattice arguments with a cyclic-extension mechanism whose nonzero carry obstructs reversal. A short lower-central calculation handles all odd primes in the small-order problem, and an invariant cyclic kernel transfers chirality in the previously studied lift. The regular and chiral applications share the full module calculations; each retains its own exact source scope and classical inputs.

1 Marked groups and intersections

We use $x^g = g^{-1}xg$ and $[x, y] = x^{-1}y^{-1}xy$. A rank-three string marking of a finite group G is a generating triple of distinct nonidentity involutions r_0, r_1, r_2 whose endpoints commute. Write $G_I = \langle r_i : i \in I\rangle$ for $I \subseteq \{0, 1, 2\}$. The marking is a *string C-group* if

$$G_I \cap G_J = G_{I\cap J} \qquad (I, J \subseteq \{0, 1, 2\}).$$

Its type is $\{k_1, k_2\}$, where $k_1 = |r_0r_1|$ and $k_2 = |r_1r_2|$. We call it nondegenerate when $k_1, k_2 \geq 3$, and tight when $|G| = 2k_1k_2$. A marked isomorphism preserves the three generators; duality reverses them. We retain the classical correspondence between string C-groups and abstract regular polyhedra when using geometric terminology.

Lemma 1 (Rank-three intersection test)**.** *A nondegenerate rank-three string marking is a string C-group if and only if*

$$\langle r_0, r_1\rangle \cap \langle r_1, r_2\rangle = \langle r_1\rangle. \tag{1}$$

In that case $2k_1k_2 \leq |G|$.

Proof. Necessity is part of the intersection property. For sufficiency set $H_{ij} = \langle r_i, r_j\rangle$. The endpoint group consists of $1, r_0, r_2, r_0r_2$. If its intersection with H_{01} contains r_2 or r_0r_2, then $r_2 \in H_{01}\cap H_{12} = \langle r_1\rangle$, contrary to distinctness. Hence $H_{02} \cap H_{01} = \langle r_0\rangle$; dually $H_{02} \cap H_{12} = \langle r_2\rangle$. These equalities and (1) also exclude a generator from the group on the other two. Singleton intersections follow, and pairs involving the empty or full index set are immediate. This accounts for every pair of subsets. Finally $|H_{01}| = 2k_1$ and $|H_{12}| = 2k_2$, so

$$|H_{01}H_{12}| = \frac{|H_{01}|\,|H_{12}|}{|H_{01} \cap H_{12}|} = 2k_1k_2 \leq |G|.$$

□

Lemma 2 (Prime-index intersection)**.** *Let C, H, J be subgroups of a group, with $C \leq H \cap J$ and $[H : C] = p$ prime. If $H \not\leq J$, then $H \cap J = C$. In particular, this holds if $\langle H, J\rangle = G$ and $J < G$.*

Proof. The index tower gives $p = [H : H \cap J][H \cap J : C]$. The first factor is not one, so the second is one. In the final situation $H \leq J$ would give $G = \langle H, J\rangle = J$, contrary to the hypothesis. □

For later products it is useful to record the elementary identities

$$(sh)^2 = 1, \qquad (sh)(bh) = sb^{-1}, \tag{2}$$

whenever $h^2 = 1$, $hsh = s^{-1}$ and $hbh = b^{-1}$. The first equality uses only the condition involving s; the second uses only the condition involving b.

2 An inverted orbit and its characters

Lemma 3 (One inverted orbit)**.** *Suppose $G = P \rtimes K$, where P has odd order and $K = \langle a, b\rangle$ is a Klein four group. Suppose $G = \langle K, r\rangle$, where r is a nonidentity involution. Write $r = ut$ with $u \in P$, $t \in K$. Then $t \neq 1$, $u^t = u^{-1}$, and*

$$P = \langle u, u^k\rangle \qquad \text{for every } k \in K \setminus \langle t\rangle.$$

Proof. An odd-order group has no nonidentity involution, so $t \neq 1$. The equation $(ut)^2 = 1$ gives $u^t = u^{-1}$. Put $L = \langle u, u^k\rangle$. Since t commutes with k, it inverts both displayed generators of L. The element k exchanges them. Thus L is K-invariant and LK is a subgroup containing K and r. It equals G, and intersecting with P gives $L = P$. □

This proof says that t inverts the two indicated generators. It does not assert that inversion is an automorphism of a nonabelian group.

Lemma 4 (Two cyclic characters)**.** *In Lemma 3, suppose P is abelian and noncyclic. Then it is the direct product of two nontrivial cyclic subgroups P_χ, indexed by the two characters $\chi : K \longrightarrow \{1, -1\}$ with $\chi(t) = -1$. The projection of u generates each P_χ.*

Proof. Write P additively. Multiplication by four is an automorphism of P. The four endomorphisms

$$e_{\epsilon,\delta} = \tfrac{1}{4}(1 + \epsilon a)(1 + \delta b), \qquad \epsilon, \delta \in \{1, -1\},$$

are orthogonal idempotents with sum the identity: expand the products using $a^2 = b^2 = 1$ and $ab = ba$. Their images give the four character components, on which a, b act by ϵ, δ respectively. The element t inverts u and commutes with K, so it inverts the whole K-orbit of u. That orbit generates P by Lemma 3; hence t acts as -1 on P. Every component with $\chi(t) = 1$ is therefore annihilated by two and is zero. For either remaining character, the projection of each orbit element is plus or minus $e_\chi u$. Its component is consequently cyclic. If one component were zero then P would be cyclic, so both are nontrivial. □

3 The complete cyclotomic lattice

Let p be an odd prime, let ζ be a primitive pth root of unity, and put $O = \mathbb{Z}[\zeta]$ and $\pi = \zeta - 1$.

Theorem 5 (Cyclotomic coordinates)**.** *For every $n \geq 1$, the ring $A_n = O/\pi^n O$ has p^n elements. Write uniquely*

$$n = (e-1)(p-1) + r, \qquad e \geq 1, \quad 1 \leq r \leq p-1.$$

As an additive group with the indicated ordered generators,

$$A_n = \langle 1\rangle \oplus \langle \pi\rangle \oplus \cdots \oplus \langle \pi^{p-2}\rangle \cong (\mathbb{Z}/p^e\mathbb{Z})^r \oplus (\mathbb{Z}/p^{e-1}\mathbb{Z})^{p-1-r}. \tag{3}$$

A modulus-one factor is trivial. Complex conjugation induces an involutory automorphism of A_n. Multiplication by ζ has order p when $n \geq 2$, and for $j \not\equiv 0 \pmod p$ it satisfies $\sum_{i=0}^{p-1} \zeta^{ij} = 0$.

Proof. The polynomial

$$\Phi_p(1+X) = p + \binom{p}{2}X + \cdots + \binom{p}{p-1}X^{p-2} + X^{p-1}$$

is Eisenstein at p. Thus O is an integral domain with $\mathbb{Z}$-basis $1, \pi, \ldots, \pi^{p-2}$. Substitution $\zeta = 1$ gives $O/\pi O \cong \mathbb{F}_p$. Multiplication by π^j induces an isomorphism $O/\pi O \longrightarrow \pi^j O/\pi^{j+1}O$: surjectivity is immediate and injectivity follows by cancellation in the domain. The successive indices are all p, whence

$$|O/\pi^n O| = p^n. \tag{4}$$

The identity $p = \prod_{j=1}^{p-1}(1-\zeta^j)$ gives $pO = \pi^{p-1}O$. Here equality, rather than inclusion alone, follows because each ratio $(1-\zeta^j)/(1-\zeta)$ is a unit of O. It belongs to O by a geometric sum. The automorphism $\zeta \mapsto \zeta^j$ identifies the norms of numerator and denominator, both equal to p, so the ratio has norm one. Its multiplication matrix on the integral basis has determinant one; the adjugate gives an integral inverse and hence a unit of O.

For $1 \le r \le p-1$ define the integral lattice

$$L_r = p\mathbb{Z}\,1 + \cdots + p\mathbb{Z}\,\pi^{r-1} + \mathbb{Z}\,\pi^r + \cdots + \mathbb{Z}\,\pi^{p-2}.$$

Every displayed generator is in $\pi^r O$, using $pO = \pi^{p-1}O$ for the first group of generators. Both L_r and $\pi^r O$ have index p^r in O, the latter by (4); thus $L_r = \pi^r O$. Now $\pi^n O = p^{e-1}\pi^r O$, which gives every coordinate modulus in (3). In particular it gives the individual orders, including trivial listed generators when $e = 1$.

The equation $\overline{\pi} = -\zeta^{-1}\pi$ shows that conjugation preserves $\pi^n O$. Multiplication by ζ has pth power the identity. For $n \ge 2$ it is not the identity, since its difference from the identity sends 1 to $\pi \notin \pi^n O$; the latter follows from the strict successive indices above. Its order is therefore p. Finally $1 + \zeta + \cdots + \zeta^{p-1} = 0$; multiplication of exponents by a nonzero residue j permutes these terms. □

Corollary 6 (Conjugation and displacement)**.** *Let A_n be written additively and let an element β of order p act by $v^\beta = \zeta v$. Put $s_i = \pi^{i-1}$ for $i \ge 1$. Then $[s_i, \beta] = s_{i+1}$ in $A_n \rtimes \langle\beta\rangle$, and*

$$\overline{s_i} = (-1)^{i-1}\zeta^{1-i}s_i, \qquad i \ge 1, \tag{5}$$

$$\zeta^{1-i}s_i = \sum_{j=i}^{p+1}\binom{p+1-i}{j-i}s_j, \qquad 1 \le i \le p. \tag{6}$$

The first expression on the left of the commutator identity is group notation; the two displayed equations are additive ring identities.

Proof. The commutator is $(\zeta - 1)s_i = \pi s_i$. Equation (5) follows by raising $\overline{\pi} = -\zeta^{-1}\pi$ to the power $i-1$. For (6), use $\zeta^{1-i} = \zeta^{p+1-i}$ and expand $(1+\pi)^{p+1-i}\pi^{i-1}$. No independence of the additional s_p, s_{p+1} is asserted or needed. □

4 Rotation pairs and reversal

Suppose $G = \langle a, b\rangle$, with $|a|, |b| \ge 3$, $(ab)^2 = 1$ and $\langle a\rangle \cap \langle b\rangle = 1$. If an involutory automorphism δ inverts both generators, form $R = G \rtimes \langle\delta\rangle$ and put

$$r_0 = a\delta, \qquad r_1 = \delta, \qquad r_2 = \delta b.$$

These are three distinct nonidentity involutions, the endpoints commute, and they generate R. The adjacent subgroups are $\langle a\rangle \rtimes \langle\delta\rangle$ and $\langle b\rangle \rtimes \langle\delta\rangle$; their intersection is exactly $\langle\delta\rangle$ by the disjoint cyclic

subgroups in G. The complete rank-three intersection test of Topic F proves the string C-group property. Thus R is the full regular group with the asserted type and order $2|G|$. This gives the complete group-theoretic regular-doubling interface; the classical string C-group–polytope correspondence is retained.

Lemma 7 (Cyclic quotient criterion)**.** *Let $\phi : G \to H$ be a homomorphism and let $a, b \in G$. If $\langle\phi(a)\rangle \cap \langle\phi(b)\rangle = 1$ and the restriction of ϕ to either $\langle a\rangle$ or $\langle b\rangle$ is injective, then $\langle a\rangle \cap \langle b\rangle = 1$.*

Proof. Any common element maps into the trivial intersection downstairs. It is therefore the identity by injectivity on the specified cyclic subgroup. This is the quotient criterion used in source Proposition 2.1; no finite-order or surjectivity hypothesis is needed for this step. □

Lemma 8 (Abelian kernel reversal)**.** *Suppose $G = P \rtimes \langle t\rangle$, where P is abelian and $t^2 = 1$. If $a \in P$ and $b = a^{-1}t$, an involutory automorphism of G simultaneously inverts a, b.*

Proof. Inversion on P is an automorphism and commutes with the action of t, so extends to an automorphism ι fixing t. Follow it by conjugation with a. The result sends a to a^{-1} and t to $a^{-1}ta$, hence sends $b = a^{-1}t$ to $a(a^{-1}ta) = ta = b^{-1}$. Its square fixes a and every element of P, and fixes t because $\iota(a) = a^{-1}$; thus it is involutory. No presentation-order argument is needed. □

5 A shared split cyclotomic construction

Theorem 9 (The cyclotomic polyhedra)**.** *For every odd prime p and every $m \geq 3$, there is a string C-group of order $4p^m$ and type $\{p, 2p\}$ with nonabelian Sylow p-subgroup. More precisely, let*

$$O = \mathbb{Z}[\zeta_p], \quad \pi = \zeta_p - 1, \quad A = O/\pi^{m-1}O, \quad P = A \rtimes \langle\beta\rangle, \qquad |\beta| = p, \quad v^\beta = \zeta_p v.$$

Define automorphisms by

$$\sigma(v) = \overline{v}, \quad \sigma(\beta) = \beta^{-1}, \qquad \tau(v) = -v, \quad \tau(\beta) = \beta.$$

Then $G = P \rtimes \langle\sigma, \tau\rangle$ has the required marking

$$r_0 = s\tau\sigma, \qquad r_1 = \beta\tau\sigma, \qquad r_2 = \sigma,$$

where s denotes the additive class of the ring element 1. It is not the identity of the additive group A.

Proof. Set $n = m - 1 \geq 2$ and write $\zeta = \zeta_p$. To fix the action convention, use pairs $(v, k) \in A \times \mathbb{Z}/p\mathbb{Z}$ with

$$(v, k)(w, l) = (v + \zeta^{-k}w, k + l). \tag{7}$$

Then $\beta = (0, 1)$ has $v^\beta = \zeta v$. The cyclotomic coordinates of Theorem 5 give $|P| = p^{n+1} = p^m$. The action by ζ is nontrivial for $n \geq 2$, so P is nonabelian. Every element outside A has order exactly p: for $k \neq 0$ its pth power has first coordinate $\sum_{i=0}^{p-1} \zeta^{-ik}v = 0$, and its image in P/A has order p.

On pairs the proposed maps are $\sigma(v, k) = (\overline{v}, -k)$ and $\tau(v, k) = (-v, k)$. They preserve (7), since conjugation sends ζ^{-k} to ζ^k. They are commuting involutory automorphisms. Moreover, σ inverts the nontrivial element β, whereas τ fixes it; and τ is nontrivial on the nonzero odd-order group A. Thus they are independent and generate a Klein four group. In particular $|G| = 4p^m$.

Put $h = \tau\sigma$. In multiplicative group notation $s^h = s^{-1}$ and $\beta^h = \beta^{-1}$. The elementary reflection identities above make $r_0 = sh$, $r_1 = \beta h$ and $r_2 = \sigma$ involutions. Directly,

$$r_0r_2 = s\tau = r_2r_0, \qquad r_0r_1 = s\beta^{-1}, \qquad r_1r_2 = \beta\tau.$$

The middle product is outside A and has order p. The last has order $2p$, since its two factors commute and have coprime orders. The marked generators are nonidentity and distinct; the first two project to $\tau\sigma$ in the complement, the third to σ, and the first two are distinct because $s \neq \beta$.

They generate the whole group. Indeed, $(r_1r_2)^p = \tau$ yields τ, then β; the generator r_2 gives σ, and r_0 gives s. The β-conjugates of s are $1, \zeta, \dots, \zeta^{p-1}$, whose classes additively generate A.

Finally $H = \langle r_0, r_1\rangle$ and $J = \langle r_1, r_2\rangle$ have orders $2p$ and $4p$. The subgroup $C = \langle r_1\rangle$ has prime index p in H. If $H \le J$, all three generators would lie in J, giving $G = J$ and contradicting $4p^m > 4p$. The prime-index lemma above therefore gives $H \cap J = C$. Its rank-three test proves the full intersection property. □

6 A nonzero cyclic carry

Theorem 10 (A nonzero carry obstructs reversal)**.** *Let p be an odd prime and let M be a finite abelian p-group. Suppose $T, S \in \mathrm{Aut}(M)$ and $s, z \in M$ satisfy*

$$T^p = S^2 = 1, \quad STS = T^{-1}, \quad Ss = s, \quad Tz = z, \quad Sz = -z, \quad |z| = p, \quad M = \langle T^i s : i \in \mathbb{Z}\rangle.$$

Put $w = z + \sum_{i=0}^{p-1} T^i s$, and suppose $|w| = p$ and $\langle w\rangle \cap \langle s\rangle = 0$. There is a group P of order $p|M|$ containing M, generated by M and β, with

$$v^\beta = Tv \ (v \in M), \qquad \beta^p = z.$$

The assignment $v \mapsto Sv$, $\beta \mapsto \beta^{-1}$ is an involutory automorphism σ of P. In $G = P \rtimes \langle\sigma\rangle$,

$$a = \beta s, \qquad b = s^{-1}\sigma$$

generate G, have orders $p^2, 2|s|$, and satisfy $(ab)^2 = 1$ and $\langle a\rangle \cap \langle b\rangle = 1$. No automorphism of G simultaneously inverts a, b.

Proof. On $M\times\mathbb{Z}$ use multiplication $(u, i)(v, j) = (u+T^{-i}v, i+j)$. The element $(-z, p)$ is central: $T^p = 1$ and $Tz = z$. Quotient by its infinite cyclic subgroup. The map $M \to P$ is injective, since an element $(-kz, kp)$ has second coordinate zero only when $k = 0$. Every coset has a unique representative (u, i) with $0 \le i < p$. This proves the order, the action and the carry relation. The map $(u, i) \mapsto (Su, -i)$ is an involutory automorphism before taking the quotient, because $ST^{-i} = T^iS$. It takes $(-z, p)$ to its inverse, so descends as stated. It is nonidentity: its induced action on $P/M \cong C_p$ is inversion.

Since $b^2 = -2s$, oddness gives $s \in \langle b\rangle$. We then recover $\beta = as^{-1}$ and $\sigma = sb$. Their conjugates recover the whole T-orbit of s, hence G. The product $ab = \beta\sigma$ is a nonidentity involution, and $|b| = 2|s|$ because s is fixed by σ. The cyclic-extension multiplication gives $a^p = w$; thus $|a| = p^2$. Any common element of $\langle a\rangle$ and $\langle b\rangle$ lies in P, hence in $\langle b^2\rangle = \langle s\rangle$. Its image in P/M is zero, so it lies in $\langle a^p\rangle = \langle w\rangle$. The assumed intersection is zero.

Suppose $\alpha \in \mathrm{Aut}(G)$ inverted a, b. The unique Sylow p-subgroup P is characteristic. From $b^2 = -2s$ we obtain $\alpha(s) = -s$. From $a = \beta s$ we obtain $\alpha(\beta) = s^{-1}\beta^{-1}s$ in multiplicative notation. Follow $\alpha|_P$ by conjugation with s^{-1}, then by σ. The resulting automorphism τ fixes β and sends s to s^{-1}. It therefore inverts every β-conjugate of s. These generate the abelian group M, so $\tau|_M = -1$. But $\beta^p = z$ gives both $\tau(z) = z$ and $\tau(z) = -z$, contradicting $|z| = p > 2$. □

The rank-three rotation-group correspondence now constructs a chiral abstract polyhedron with these generators. That geometric correspondence is retained explicitly; the preceding argument proves the group, generation, intersection and failure of reversal themselves.

7 Class-three collection and small kernels

In a group of class at most three put $c = [x, y]$, $d = [c, x]$ and $e = [c, y]$. The elements d, e are central. The commutator identities $[uv, w] = [u, w]^v[v, w]$ and $[u, vw] = [u, w][u, v]^w$ give by induction

$$[x^k, y] = c^k d^{\binom{k}{2}}, \qquad [x, y^k] = c^k e^{\binom{k}{2}} \quad (k \ge 0).$$

Collecting all x factors first, then y, c, d, e, similarly gives

$$(xy)^k = x^k y^k c^{-\binom{k}{2}} d^{-\binom{k}{3}} e^{-2\binom{k}{3}-\binom{k}{2}}.$$

For completeness, if C_k, D_k, E_k are the last three exponents, moving the final x, y to their positions gives $C_{k+1} = C_k - k$, $D_{k+1} = D_k + C_k$ and $E_{k+1} = E_k + C_k - k - \binom{k}{2}$, with all three initial values zero. These recurrences give exactly the displayed formula. In particular, when $p \geq 5$ and the commutator subgroup has exponent p, all correction factors disappear in $(xy)^p$. Weight-three commutators are multiplicative in each variable: expand the same two commutator identities; every correction has weight four and vanishes. Thus simultaneous inversion of the first two variables inverts each of $[x, y, x]$ and $[x, y, y]$. Also $[x^{-1}, y^{-1}]^{xy} = [x, y]$, so the new weight-two commutator has the same order.

Lemma 11. *Let p be an odd prime. If a nonabelian two-generated group P has order p^3 or p^4, then P' has exponent p and the nilpotency class is at most three. If the class is three, then*

$$|P'| = p^2, \quad Z(P) = \gamma_3(P) \cong C_p, \quad P/P' \cong C_p^2.$$

Proof. A nonabelian finite p-group has noncyclic abelianization. For example, if P/P' were cyclic, its elementary abelian Frattini quotient would be cyclic and the Burnside basis theorem would make P cyclic. Thus $|P'| \leq p^2$. Every nontrivial step in the lower central series strictly decreases order, giving class at most three. In class two, if P' had order and exponent p^2, then $|Z(P)| \geq p^2$ and $P/Z(P)$ would have order at most p^2. It cannot be cyclic, so is C_p^2; hence $x^p \in Z(P)$ for all x, and $[x, y]^p = [x^p, y] = 1$. Since a two-generator class-two commutator group is cyclic, this is a contradiction.

In class three necessarily $|P| = p^4$, $|P'| = p^2$, $|\gamma_3(P)| = p$, and $P/P' = C_p^2$. Also $|Z(P)| = p$: a center of order at least p^2 would give a quotient of order at most p^2, hence an abelian quotient and class at most two. Therefore $Z(P) = \gamma_3(P)$. Suppose P' were cyclic of order p^2. For a generating pair x, y, write $c = [x, y]$. Its image generates $P'/\gamma_3(P)$, so c generates P'. Write

$$c^x = c^{1+\lambda p}, \quad c^y = c^{1+\mu p}, \quad x^p = c^a, \quad y^p = c^b.$$

The conjugation exponents have the displayed shape because the conjugation action is a p-group. Class-three collection and $p \mid \binom{p}{2}$ give $[x^p, y] = c^p$ and $[x, y^p] = c^p$; all correction terms are powers of c^{p^2}. Comparison with the displayed powers yields $a\mu \equiv 1$ and $-b\lambda \equiv 1 \pmod{p}$. But $[x^p, x] = 1$ gives $a\lambda \equiv 0 \pmod{p}$, a contradiction. Thus $P' \cong C_p^2$, proving the assertion. □

8 The previously proved cyclic lift

Theorem 12 (A cyclic lift with an invariant kernel)**.** *Let p be odd and let $(G_0; a_0, b_0)$ be a chiral rotation pair with $|a_0| = p^2$, $|b_0| = 2p$. Suppose a surjection $\phi : G_0 \to C_p = \langle c \rangle$ sends a_0, b_0 to c, c^{-1}. For $C_{p^2} = \langle u \rangle$ the subgroup generated by $a = (a_0, u)$ and $b = (b_0, u^{-1})$ in $G_0 \times C_{p^2}$ is a chiral rotation group of order $p|G_0|$ and type $\{p^2, 2p^2\}$.*

Proof. The two generators lie in the fibre product $F = \{(g, u^j) : \phi(g) = c^j\}$. Projection has kernel $\langle (1, u^p) \rangle$, of order p. The element $b^{2p} = (1, u^{-2p})$ generates that kernel, and the marked subgroup projects onto G_0. It is therefore F, of order $p|G_0|$. The direct-product orders give the stated types and $|ab| = 2$. Projection is injective on $\langle a \rangle$, since its image has the same order p^2. A common element of $\langle a \rangle$ and $\langle b \rangle$ maps to one, hence is one. Finally any simultaneous reversal preserves the explicitly identified kernel $\langle b^{2p} \rangle$ and induces a reversal on G_0, a contradiction. This argument does not assume that arbitrary chiral covers stay chiral, nor that this kernel is characteristic under every automorphism. □

This is the earlier central-cyclic-lift proof, included in full with its original scope. Its use below carries no new question or historical-priority claim.

9 Attribution and formal scope

The first three sections retain the accepted Topic F arguments and attribution: the rank-three test and Klein-four orbit structure reorganize Hou, Feng, Leemans and Qu, and the integral cyclotomic arithmetic is classical. The complete split construction, previously in the first application, is shared here because the second application uses two of its regular families. The cyclic-carry theorem reorganizes the constructions of Kong, Feng, Hou, Leemans and Qu. The small-kernel lemma replaces the classification and finite small-prime lookup in their tightness proof; the earlier cyclic lift is included with its exact kernel hypothesis. No historical priority claim is made for these results.

The classical regular and chiral group-to-polyhedron correspondences, Burnside's basis theorem and elementary finite-group facts remain explicit background. The applications separately identify their retained classification inputs; none is inferred from a finite sample. Five unchanged declarations in `OddKernelIntersection` cover prime-index intersections and reflection identities. Four unchanged `LiftCertificate` declarations cover kernel/intersection transport (three are used by the cyclic lift). Three new `CyclicCarryObstruction` declarations prove inversion on an abelian generated subgroup and the fixed-power/inverted-carry contradiction. The ring, group constructions, collection identities, small-order theorem and geometric correspondences remain written, not formally certified application instances.

References

[1] D.-D. Hou, Y.-Q. Feng, D. Leemans and H.-P. Qu, *String C-groups of order* $4p^m$, arXiv:2407.10388v1, 2024.

[2] T.-T. Kong, Y.-Q. Feng, D.-D. Hou, D. Leemans and H.-P. Qu, *Chiral 3-polytopes of order* $2p^m$, 22-page accepted manuscript, 19 January 2026, DOI10.26493/1855-3974.3716.b64.